Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for retail and service businesses, turning owners' growth goals into site-selection analyses and revenue targets; identified a $1.5M expansion opportunity.
- Built a revenue driver tree (traffic × capture rate × ticket price, with a traffic-to-capture elasticity) and a margin model (fixed and variable costs per car) that translated client profit goals into break-even revenue and volume targets.
- Estimated the incremental revenue of new locations with Double ML and causal forests (econml, XGBoost nuisance models) on a 12-year county panel, stating identification assumptions and checking robustness with difference-in-differences and a synthetic-data recovery test.
- Back-tested demand predictions out of sample against real operating sites, measuring bias, error, rank correlation, and 90% interval coverage before using the model for planning.
- Built low/base/high scenarios and Monte Carlo risk models (probability of loss, VaR, CVaR) that carry the causal estimates' 95% confidence intervals, and explained sensitivity and uncertainty to owners in written reports and maps.
- Designed a site-scoring model combining demographic, traffic, competitive-density, and consumer-behavior features to rank locations by revenue potential and risk.
- Engineered reproducible Python pipelines with data-quality checks and 150+ unit tests on Census, traffic, and competitor data, cutting analysis turnaround by 91% (weeks to days), and documented each method in reusable walkthroughs.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Led the healthcare marketing analytics and data science roadmap with strategy, finance, engineering, and behavioral science leadership, owning the KPI framework that connected acquisition (gross lead rate, GLR) to plan enrollment and active membership (member rate) and acquisition cost (cost per member, CAC).
- Redefined GLR for a new client whose data lacked key lead markers: isolated the missing fields with data engineering and set one definition (a lead counts on the date it came in, is credited to the last response, and is flagged when broker-influenced) so GLR was reported accurately.
- Partnered with data engineering on curated Snowflake and Databricks tables that served as the shared source for GLR, member rate, and cost per member across models and reporting.
- Built a causal graph (DAG) from historical campaign data showing GLR drives member rate, which drives cost per member, focusing model building and causal tests on GLR as the lever.
- Built an in-house experimentation tool and standardized cross-team A/B testing and causal inference methods, which data science and strategy used to report results to clients, validate lift across segmented prospect audiences, and set each campaign's ad-spend budget; CUPED cut metric variance by 20%.
- Built predictive, segmentation, and uplift models on millions of HIPAA-protected (PII/PHI) Medicare/Medicaid-eligible records, used in 87% of campaigns, targeting incremental (causal) response rather than predicted response and cutting ad spend by $10K per campaign.
- Identified one client's "Opportunist" prospect segment as the main driver of GLR, shifting campaign targeting toward that audience.
- Linked the monthly retention survey and member funnel to retention behavior, finding that claims and ticket complaints were the main drivers of churn, with driver and segment breakdowns in monthly reporting.
- Owned the monthly model monitoring report on drift, performance, and media mix forecast-vs-actual accuracy for up to 10 production models (media mix, response, eligibility, psychology-match), read by the directors of data science and strategy and the managing partner.
- Maintained a company-wide self-service dashboard of model metrics, refreshed by scheduled Databricks jobs that also fed the retraining and scoring pipelines.
- Re-architected a production churn prediction platform in Python, SQL, Spark, Snowflake, and Databricks, reducing runtime by 48% and memory use by 23%.

Founding Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined success metrics, evaluation frameworks, and drift monitoring for production search and pricing models, using offline validation and online experiments, and mentored the team on ML practices.
- Built SQL training tables that broke bulk orders into item-level shipping costs, then shipped a regression pricing API (C#, Azure) with engineering, scaling to 10,000+ daily requests.
- Reduced customer dissatisfaction with search by 12% and drove a 16% increase in platform engagement with product and engineering partners through custom Elasticsearch indexing, tokenizers, and TF-IDF weighting.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Defined and reported the core campaign metrics (GLR, member rate, cost per member) used to evaluate client campaigns.
- Optimized top-of-funnel GLR to accelerate pipeline velocity, driving a 24% increase in paid member conversions within an average 60-day time horizon.
- Built an out-of-sample leading indicator test showing GLR changes predict member rate shifts 60 days in advance ($R^2 = 0.28$).
- Built and validated XGBoost classification models on millions of consumer records with hundreds of external attributes, driving a 3% lift in response rates, and explained model drivers to stakeholders with SHAP.
- Automated reporting pipelines and Power BI dashboards with Python, SQL, and Snowflake, removing 60 minutes of manual work daily.
- Improved geospatial targeting accuracy by 20% through GIS analysis with GeoPandas, focusing campaign resources on high-opportunity regions.

Graduate Researcher & Teaching Assistant — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in mathematical optimization, functional analysis, and operator theory for doctoral dissertation research.
- Taught mathematics and statistics to classes of 75-200 students and mentored 100+ students in summer programs, achieving a 95% pass rate.

TECHNICAL PROJECTS

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes to show when the LTV = ARPU / churn identity breaks down (tenure-dependent churn), and estimated lifetime value from empirical survival curves instead.

Mobile Game Retention A/B Test github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Ran a full experiment workflow on 1- and 7-day player retention: hypotheses, outlier handling, sample sizing, significance testing, and practical significance for the launch call.

Medicare Claims Intelligence & Fraud Detection github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a tested dbt warehouse (staging to marts) over 790K CMS Medicare claims and a provider fraud (FWA) detector that flagged planted billing schemes with 81% precision.
- Cut forecast error 41% vs. a seasonal benchmark in rolling backtests (ARIMA, ETS, Prophet), built a SHAP-explained XGBoost model that finds 46% more high-cost members than baseline (MLflow), and a RAG assistant with PHI guardrails.

Starbucks Rewards: Predicting Offer Responses github.com/beauvilerobed/predict-consumer-responses

- Predicted how rewards app members respond to BOGO, discount, and informational offers from transaction, offer, and demographic data, with an XGBoost multi-class model trained and deployed on AWS SageMaker.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Metrics & Planning: KPI Trees & Driver Decomposition, Metric Definitions, Acquisition Funnel & CAC, Retention & Cohort Analysis, Survival Analysis, LTV / ARPU, Sensitivity & Scenario Analysis, Target Setting, Forecast Validation (Backtesting)

Causal Inference & Experimentation: A/B Testing, CUPED, Power Analysis, Double ML, Causal Forests, Difference-in-Differences, CATE / Uplift Modeling, Robustness Checks, Monte Carlo Simulation

Modeling: Regression, Classification, XGBoost, Time Series Forecasting (ARIMA, ETS, Prophet), Media Mix Modeling, Causal Graphs (DAGs), Graph Mining (NetworkX), SHAP, Model Monitoring & Drift

Data & Tools: SQL, Python (pandas, NumPy, statsmodels, scikit-learn, econml), dbt, Snowflake, Spark, Databricks, SQL Server, PostgreSQL, AWS, Azure, Curated (Gold) Tables, Git, Unit Testing, AI-Assisted Development (Claude Code)

Dashboards & Communication: Power BI, Tableau, Excel, Plotly, Self-Service Dashboards, Method Documentation, Stakeholder & Client Presentations, Mentoring